TAMIL NADU STATE BOARD - STANDARD 11 CHEMISTRY VOLUME 2

CHAPTER 14: HALOALKANES AND HALOARENES

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Chemistry Volume 2
Chapter Name:	Haloalkanes and Haloarenes

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)

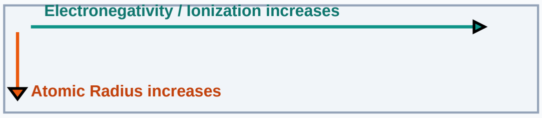
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
SN1 Reaction	A nucleophilic substitution reaction occurring in two steps, whose rate depends only on alkyl halide concentration.
SN2 Reaction	A nucleophilic substitution reaction occurring in a single concerted step, leading to Walden inversion of configuration.
Fittig Reaction	The coupling of two aryl halides with sodium metal in dry ether to form diphenyl.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

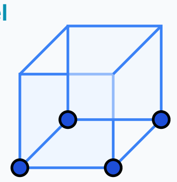
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Simple Cubic Lattice System

Essential Formulas, Laws & Identities:

- SN1 Rate: Rate = k * [R-X] (forms carbocation intermediate)
- SN2 Rate: Rate = k * [R-X] * [Nu-] (backside attack)

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

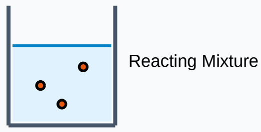
Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Why is chlorobenzene less reactive towards nucleophilic substitution than chloroethane?

Step-by-step Solution:
In chlorobenzene, the lone pair on chlorine enters into resonance with the benzene ring, imparting partial double bond character to C-Cl bond, making it stronger and harder to cleave.

Titration Beaker & Solute Precipitation



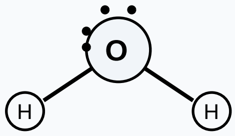
HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Contrast SN1 and SN2 reaction pathways regarding steps, kinetics, intermediates, and stereochemical outcomes.

Analytical Solution Strategy:
SN1: 2 steps, 1st order kinetics, carbocation intermediate, racemization. Favored by polar solvents and tertiary halides.
SN2: 1 step, 2nd order kinetics, transition state, inversion. Favored by primary halides and strong nucleophiles.

Lewis Dot Structure (H_2O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State Swarts reaction and Finkelstein reaction equations.

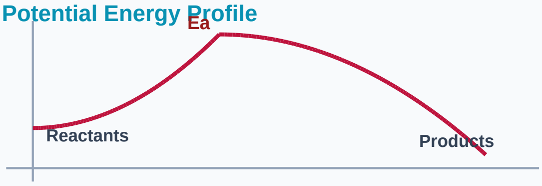
Q2 (2-Mark): Write a note on Sandmeyer's reaction.

Q3 (2-Mark): What are Grignard reagents? How are they prepared?

Q4 (5-Mark): Explain elimination reactions in haloalkanes (Saytzeff's Rule).

Q5 (5-Mark): Why are haloalkanes polar in nature?

Chemical Reaction Potential Energy Profile



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Anesthetics

Halothane gas is inhaled to act as a general anesthetic during clinical surgeries.

Application Case: Agriculture

Organochlorine compounds act as heavy insecticides protecting bulk food crops.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C_6H_6) Resonance

Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

Electrochemical Galvanic Cell System